



DaVinci 200

Preventive Maintenance Checklist

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1.1	preReleas	25.03.2022	DW	Odoo sparepart numbers adjustment for LWM200 Systmes
1.2	preReleas	08.01.2025	DW / SAz	Safety check list and needed changes discussed for LP checks and stage
1.3	Release	09.01.2025	SAz	Discussed changes integrated
1.4	Release	10.01.2025	HL	Remove exchange ionizer tips (3.2)
1.5	Release	22.05.2025	EL / HL	Consolidate other DaVinci info
1.6	Release	02.10.2025	EL	Copy of 150OC. Update Main image & Tool Name. Add name of safety-document

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1 Introduction

1.1 Maintenance duration

The annual maintenance takes approx. 1 day.

1.2 Preparation

To carry out maintenance successfully, certain components and materials must be prepared. These components and materials are listed in chapter 3.

1.3 Execution

Maintenance is carried out according to the list in chapter 2.

2 Checklist

2.1 Robot Check

Task	Interval			Done
	Every 3 Years	Every 2 Years	Every year	
Check robot paddle:				
• Check the paddles for damage			X	☐
• Check the vacuum pads for damage			X	☐
Check robot vacuum:				
• Check the vacuum pressure on robot/robot paddles			X	☐
• Check the vacuum valve			X	☐
• Visual inspection of the vacuum lines for damage			X	☐
Check robot positions:				
• Loadport 1			X	☐
• Loadport 2			X	☐
• Loadport 3 (If installed)			X	☐
• Process Module 1			X	☐
• Process Module 2 (If installed)			X	☐
Visual inspection				
• Check that the robot runs smoothly			X	☐
• Check for damage to the robot			X	☐

2.2 Loadport

Task	Interval			Done
	Every 3 Years	Every 2 Years	Every year	
Visual inspection				
• Check the Loadports for damage on the housing			X	☐
• Check the support points for the carrier			X	☐
• Check the sensors for damage			X	☐
• Carrier Size Detection Test (if installed)			X	☐
Manual moving				
• Load the Loadports manually by software (if available e.g. ToolComan- der) and check for noise			X	☐
• Unload the Loadports manually by software (if available e.g. ToolCo- mander) and check for strange noise			X	☐
• Check the flaps for opening and closing (if installed)			X	☐
• Check the mapping function with different materials			X	☐

2.3 Prealigner

Task	Interval			Done
	Every 3 Years	Every 2 Years	Every year	
Visual inspection				
• Check the prealigner for damage			X	☐
• Check the contact areas for the wafer			X	☐
Manuel moving				
• Check the alignment accuracy of the prealigner (Change the angle offset if necessary)			X	☐
• Check the vacuum on the prealigner (fetch the wafer by software)			X	☐

2.4 Barcode reader

Task	Interval			Done
	Every 3 Years	Every 2 Years	Every year	
Frontside reader (if installed)				
• Check barcode reader position			X	☐
• Check, if the reader can read the customer ID's			X	☐
Backside reader (if installed)				
• Check barcode reader position			X	☐
• Check, if the reader can read the customer ID's			X	☐

2.5 Stage

Task	Interval			Done
	Every 3 Years	Every 2 Years	Every year	
Visual inspection on the stages of process modules				
• Check the stage for damage			X	☐
• Check the chuck surface for damage, scratches, resist and particles			X	☐
• Check the cable chains for abrasion (if installed)			X	☐
• Check the stage surface for scratches			X	☐
Check mechanic and function of process modules				
• Check the smooth running of the carriage			X	☐
• Check the signal from the position measurement system (if installed)			X	☐
• Check the robot handover point (chuck load position)			X	☐
• Check the vacuum signals on vacuum controller			X	☐
• Check the vacuum signals on AIS GUI			X	☐
• Check the vacuum guides (if available)			X	☐
Clean process modules				
• Clean the stage				
• Clean the energy chains (if available)			X	☐
• Clean the guides with a dry cleanroom wipe (if available)			X	☐
• Grease the linear guides (if available)			X	☐

2.6 Optics

Task	Interval			Done
	Every 3 Years	Every 2 Years	Every year	
General Check of optic components				
Check the system for damage			X	☐
Check that no components are loose, no strong forces are recommended to these delicate parts (objectives, camera...)			X	☐
Service Macros				
Execute the Service Macro: Objective Revolver Offset (Check if the set values are correct, replace if not correct)			X	☐
Check the Service Macro: LFS-Offset (Save the values)			X	☐

2.7 Control Cabinet

Task	Interval			Done
	Every 3 Years	Every 2 Years	Every year	
Visual inspection				
• Check if components are loose (apparent damage, loose wires...)			X	☐
• Check PLC for any apparent errors			X	☐
Check fuse				
• Test the FI (if installed)			X	☐
• Test the fuses			X	☐

2.8 Housing

Task	Interval			Done
	Every 3 Years	Every 2 Years	Every year	
Visual inspection				
• Check the housing for damage			x	☐
• Check the labeling (maybe a safety label is lost)			x	☐
• Check for damage on the top side			x	☐
Check FFU				
• Check for loud noises			x	☐
Ionizer				
• Check Ionizer		x		☐
• Clean Ionizer tips			x	☐
Check IP Cams				
• Check the IP cams for performance		x		☐
• Exchange the SD card in the IP cams		x		☐

2.9 Safety Check

Task	Interval			Done
	Every 3 Years	Every 2 Years	Every year	
Safety Check				
Perform complete safety check. Follow the “DaVinci 200 A-Star_Safety-Test V1.3_EN.pdf” instruction			x	☐

3 Provision of materials

3.1 Equipment and additional documents

Task	Units			Done
	Every 3 Years	Every 2 Years	Every year	
• “DaVinci 200 A-Star_SafetyTest V1.3_EN.pdf” instruction			X	☐

3.2 Spare parts list

Task	Article Number	Units			Done
		Every 3 Years	Every 2 Years	Every year	
• Clean ionizer tips	GMK002.780	-	-	2	☐
• Exchange SD card in the IP cams	K0020527-010001 (μ1.014.460)	-	2	-	☐

3.3 Expendable materials

Task	Article Number	Units			Done
		Every 3 Years	Every 2 Years	Every year	
• Dry cleanroom wipe		X	X	X	☐
• Ladder (min. 2.0m height)		X	X	X	☐
• Cleanroom air to clean the optic		X	X	X	☐

4 Notice

A large rectangular area filled with a grid of small, evenly spaced dotted lines, intended for handwritten notes.

5 Notice

Customer: _____

Maintenance performed on: _____

Location _____

Maintenance performed by: _____

Tool: _____

Confirmation that the Preventive Maintenance was performed:

Customer: _____

MueTec Service: _____